

GONGYANG LI

Associate Professor, Shanghai University

Email: ligongyang@shu.edu.cn / lllmiemie@163.com

<https://mathlee.github.io>

MAJOR EMPLOYMENT HISTORY

- Associate Professor, Shanghai University, Shanghai, China 06/2024–Now
 - School of Communication and Information Engineering
- Postdoc, Shanghai University, Shanghai, China 06/2022–06/2024
 - Supervisor: Professor Xinpeng Zhang

EDUCATION

- Ph.D., Shanghai University, Shanghai, China 09/2016–06/2022
 - Successive Master-Doctor Program
 - Major in Signal and Information Processing
 - Advisor: Professor Zhi Liu
- Visiting Ph.D. Student, Nanyang Technological University, Singapore 07/2021–06/2022
 - Major in Computer Vision, Media & Interactive Computing Lab (MICL-CVL)
 - Advisor: Professor Weisi Lin (IEEE/IET Fellow and Honorary Fellow of SIET)
- B.S., Shanghai Normal University, Shanghai, China 09/2012–06/2016
 - Major in Communication Engineering

RESEARCH INTERESTS

Computer Vision, Saliency Detection, Image/Video Object Segmentation, Semantic Segmentation

AWARDS AND PRIZES

- National Scholarship for Doctoral Students, 2020
- President's Scholarship of Shanghai University, 2020
- The Baogang Scholarship, 2020
- The 13th China Graduate Electronic Design Competition (Shanghai Division), 2nd prize, 2018
- The 14th China Graduate Mathematical Modeling Contest, 3rd prize, 2017
- Shanghai Outstanding Graduates, 2016

PUBLICATIONS

Google Scholar profile: <https://scholar.google.com/citations?hl=zh-CN&user=YNq7jc8AAAAJ>

RGB-D Salient Object Detection

- **Gongyang Li**, Zhi Liu, Linwei Ye, Yang Wang, and Haibin Ling, “Cross-modal weighting network for RGB-D salient object detection,” *European Conference on Computer Vision (ECCV)*, 2020.
- **Gongyang Li**, Zhi Liu, and Haibin Ling, “ICNet: Information conversion network for RGB-D based salient object detection,” *IEEE Trans. on Image Processing (T-IP)*, 29:4873-4884, 2020. (IF:11.041)
- **Gongyang Li**, Zhi Liu, Minyu Chen, Zhen Bai, Weisi Lin, and Haibin Ling, “Hierarchical alternate interaction network for RGB-D salient object detection,” *IEEE Trans. on Image Processing (T-IP)*, 30:3528-3542, 2021. (IF:11.041)

- Shixiang Shi, **Gongyang Li**[✉], Runmin Cong, Shunxin Xiao, and Weisi Lin, “Diffusion-driven RGB-D salient object detection with temporal modulation,” *IEEE Trans. on Circuits and Systems for Video Technology (T-CSVT)*, 36(8):11831-11843, 2026. ([✉] corresponding author)
- Jianlin Chen, **Gongyang Li**[✉], Zhijiang Zhang[✉], Liang Chang, and Dan Zeng, “STENet: Superpixel token enhancing network for RGB-D salient object detection,” *IEEE Trans. on Multimedia (T-MM)*, 2026. ([✉] corresponding author)
- Wenqi Si, **Gongyang Li**[✉], Shixiang Shi, and Weisi Lin, “Weakly-supervised RGB-D salient object detection via SAM-driven pseudo annotation and state space interaction-based diffusion,” *IEEE Trans. on Multimedia (T-MM)*, 2026. ([✉] corresponding author)
- Xiaofei Zhou, **Gongyang Li**, Chen Gong, Zhi Liu, and Jiyong Zhang, “Attention-guided RGBD saliency detection using appearance information,” *Image and Vision Computing*, 95:103888, 2020. (IF:3.860)
- Zhen Bai, Zhi Liu, **Gongyang Li**, Linwei Ye, and Yang Wang, “Circular complement network for RGB-D salient object detection,” *Neurocomputing*, 451:95-106, 2021. (IF:5.779)

Salient Object Detection in Optical Remote Sensing Images

- **Gongyang Li**, Zhi Liu, Weisi Lin, and Haibin Ling, “Multi-content complementation network for salient object detection in optical remote sensing images,” *IEEE Trans. on Geoscience and Remote Sensing (T-GRS)*, 60:5614513, 2022. (IF:8.125)
- **Gongyang Li**, Zhi Liu, Zhen Bai, Weisi Lin, and Haibin Ling, “Lightweight salient object detection in optical remote sensing images via feature correlation,” *IEEE Trans. on Geoscience and Remote Sensing (T-GRS)*, 60:5617712, 2022. (IF:8.125)
- **Gongyang Li**, Zhi Liu, Dan Zeng, Weisi Lin, and Haibin Ling, “Adjacent context coordination network for salient object detection in optical remote sensing images,” *IEEE Trans. on Cybernetics*, 53(1):526-538, 2023. (IF:19.118)
- **Gongyang Li**, Zhi Liu, Xinpeng Zhang, and Weisi Lin, “Lightweight salient object detection in optical remote-sensing images via semantic matching and edge alignment,” *IEEE Trans. on Geoscience and Remote Sensing (T-GRS)*, 61:5601111, 2023. (IF:8.125)
- **Gongyang Li**, Zhen Bai, Zhi Liu, Xinpeng Zhang, and Haibin Ling, “Salient object detection in optical remote sensing images driven by Transformer,” *IEEE Trans. on Image Processing (T-IP)*, 32:5257-5269, 2023. (IF:11.041)
- **Gongyang Li**, Zhen Bai, and Zhi Liu, “Texture-semantic collaboration network for ORSI salient object detection,” *IEEE Trans. on Circuits and Systems II: Express Briefs (T-CAS-II)*, 71(4):2464-2468, 2024. (IF:4.4)
- **Gongyang Li**, Shixiang Shi, Yong Wu, Weisi Lin, and Zhen Bai, “Lightweight ORSI salient object detection via frequency and mutual assistance attention,” *IEEE Trans. on Geoscience and Remote Sensing (T-GRS)*, 64:5617112, 2026.
- **Gongyang Li**, Zhen Bai, Runmin Cong, Dan Zeng, Weisi Lin, and Xiao-Ping Zhang, “IPDiff: Diffusion-driven ORSI salient object detection with information reconstruction and multi-prior guidance,” *International Journal of Computer Vision (IJCV)*, 134(7):341, 2026.
- Yihua Tu, Wenqi Si, **Gongyang Li**[✉], Chengjun Han, Yun Sui, and Weisi Lin, “Exploring information entropy-driven interaction and hierarchical edge perception for lightweight ORSI salient object detection,” *IEEE Trans. on Circuits and Systems for Video Technology (T-CSVT)*, 2026. ([✉] corresponding author)
- Zhen Bai, **Gongyang Li**[✉], and Zhi Liu, “Global-local-global context-aware network for salient object detection in optical remote sensing images,” *ISPRS Journal of Photogrammetry and Remote Sensing (ISPRS JP&RS)*, 198:184-196, 2023. (IF: 11.774) ([✉] corresponding author)
- Xian Fang, Qihang Wang, Qiaohong Chen, and **Gongyang Li**, “Interactive edge awareness network for salient object detection in optical remote sensing images,” *Expert Systems With Applications (ESWA)*, 316:131879, 2026.

Salient Object Detection on Other Images/Videos

- Yilei Chen[#], **Gongyang Li**[#], Ping An, Zhi Liu, Xinpeng Huang, and Qiang Wu, “Light field salient object detection with sparse views via complementary and discriminative interaction network,” *IEEE Trans. on Circuits and Systems for Video Technology (T-CSVT)*, 34(2):1070-1085, 2024. ([#] equal contribution) (IF:5.859)
- Mengke Huang, **Gongyang Li**[✉], Zhi Liu, and Linchao Zhu, “Lightweight distortion-aware network for salient object detection in omnidirectional images,” *IEEE Trans. on Circuits and Systems for Video Technology (T-CSVT)*, 33(10):6191-6197, 2023. (IF: 5.859) ([✉] corresponding author)
- Xiaofei Zhou, Zhi Liu, Chen Gong, **Gongyang Li**, and Mengke Huang, “Video saliency detection using deep convolutional neural networks,” *PRCV*, 2018.
- Jingru Ren, Zhi Liu, **Gongyang Li**, Xiaofei Zhou, Cong Bai, and Guangling Sun, “Co-saliency detection using collaborative feature extraction and high-to-low feature integration,” *IEEE ICME*, 2020.
- Mengke Huang, Zhi Liu, **Gongyang Li**, Xiaofei Zhou, and Olivier Le Meur, “FANet: Features adaptation network for 360° omnidirectional salient object detection,” *IEEE Signal Processing Letters (SPL)*, 27:1819-1823, 2020. (IF:3.201)
- Zhen Bai, Zhi Liu, **Gongyang Li**, and Yang Wang, “Adaptive group-wise consistency network for co-saliency detection,” *IEEE Trans. on Multimedia (T-MM)*, 25:764-776, 2023. (IF:8.182)

Saliency Prediction

- Mengke Huang, **Gongyang Li**[✉], Zhi Liu, Yong Wu, Chen Gong, Linchao Zhu, and Yi Yang, “Exploring viewport features for semi-supervised saliency prediction in omnidirectional images,” *Image and Vision Computing (IVC)*, 129:104590, 2023. (IF:3.860) ([✉] corresponding author)
- Andrey Moskalenko, Alexey Bryncev, Ivan Kosmynin, Kira Shilovskaya, Mikhail Erofeev, Dmitry Vatulov, Radu Timofte, Kun Wang, Yupeng Hu, Zhiran Li, Hao Liu, Qianlong Xiang, Liqiang Nie, Konstantinos Chaldaiopoulos, Niki Efthymiou, Athanasia Zlatintsi, Panagiotis Filntisis, Katerina Pastra, Petros Maragos, Li Yang, Gen Zhan, Yiting Liao, Yabin Zhang, Yuxin Liu, Xu Wu, Yunheng Zheng, Linze Li, Kun He, Cong Wu, Xuefeng Zhu, Tianyang Xu, Xiaojun Wu, Wenzhuo Zhao, Keren Fu, **Gongyang Li**, Shixiang Shi, Jianlin Chen, Haibin Ling, Yaoxin Jiang, Guoyi Xu, Jiajia Liu, Yaokun Shi, and Jiachen Tu, “NTIRE 2026 Challenge on Video Saliency Prediction: Methods and Results,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2026.
- Ziqiang Wang, Zhi Liu, **Gongyang Li**, Yang Wang, Tianhong Zhang, Lihua Xu, and Jijun Wang, “Spatio-temporal self-attention network for video saliency prediction,” *IEEE Trans. on Multimedia (T-MM)*, 25:1161-1174, 2023. (IF:8.182)
- Yingjie Song, Zhi Liu, **Gongyang Li**, Dan Zeng, Tianhong Zhang, Lihua Xu, and Jijun Wang, “RINet: Relative importance-aware network for fixation prediction,” *IEEE Trans. on Multimedia (T-MM)*, 25:9263-9277, 2023. (IF:8.182)
- Jiawei Xie, Zhi Liu, **Gongyang Li**, Xiaofeng Lu, and Tao Chen, “Global semantic-guided network for saliency prediction,” *Knowledge-Based Systems*, 284:111279, 2024. (IF:8.8)
- Jiawei Xie, Zhi Liu, **Gongyang Li**, and Yingjie Song, “Audio-visual saliency prediction with multi-sensory perception and integration,” *Image and Vision Computing (IVC)*, 143:104955, 2024. (IF:3.860)

Defect Detection/Segmentation

- **Gongyang Li**, Xiaofei Zhou, and Hongyun Li, “Ordered cross-scale interaction network for no-service rail surface defect segmentation,” *IEEE Trans. on Instrumentation and Measurement (T-IM)*, 74:5033210, 2025. (IF:5.6)
- **Gongyang Li**[#], Chengjun Han[#], and Zhi Liu, “No-service rail surface defect segmentation via normalized attention and dual-scale interaction,” *IEEE Trans. on Instrumentation and Measurement (T-IM)*, 72:5020310, 2023. ([#] equal contribution) (IF:5.332)

- Chengjun Han[#], **Gongyang Li**[#], Zhi Liu, and Yike Wang, “Two-stage edge reuse network for salient object detection of strip steel defect images,” *IEEE Trans. on Instrumentation and Measurement (T-IM)*, 71:5019812, 2022. ([#] equal contribution) (IF:5.332)
- Tuo Ding[#], **Gongyang Li**[#], Zhi Liu, and Yike Wang, “Cross-scale edge purification network for salient object detection of steel defect images,” *Measurement*, 199:111429, 2022. ([#] equal contribution) (IF:5.131)
- Xiaofei Zhou, Xuan Wang[✉], **Gongyang Li**[✉], Deyang Liu, Qingshan She, Xiaobin Xu, and Runmin Cong, “Few-shot strip steel surface defect segmentation via pre-trained variational auto-encoder-based latent gaussian process regression,” *IEEE Trans. on Image Processing (T-IP)*, 35:2771-2786, 2026. ([✉] corresponding author)
- Jianlin Chen, **Gongyang Li**[✉], Zhijiang Zhang[✉], and Dan Zeng, “Hierarchical prior-guided and channel-wise adaptation fusion network for RGB-D railway surface defect inspection,” *Journal of Visual Communication and Image Representation (JVCI)*, 118:104832, 2026. ([✉] corresponding author)
- Zhihao He, **Gongyang Li**[✉], and Zhi Liu[✉], “RGB-D rail surface defect inspection driven by conditional diffusion architecture and frequency knowledge,” *IEEE Sensors Journal (SJ)*, 25(10):18334-18343, 2025. ([✉] corresponding author) (IF:4.3)
- Xiaofei Zhou, Zhenkun Mo, **Gongyang Li**, Liuxin Bao, Xiaobin Xu, and Jiyong Zhang, “Re-parameterization-driven depthwise separable large-kernel network for lightweight salient object detection of strip steel surface defects,” *IEEE Trans. on Cybernetics (T-CYB)*, 56(8):4269-4281, 2026.
- Zhihao He, Zhi Liu, and **Gongyang Li**, “Hierarchical cross-modal attention and multi-scale interaction for RGB-D rail surface defect inspection,” *Displays*, 2026.

Fixation-based Object Segmentation

- **Gongyang Li**, Zhi Liu, Ran Shi, Zheng Hu, Weijie Wei, Yong Wu, Mengke Huang, and Haibin Ling, “Personal fixations-based object segmentation with object localization and boundary preservation,” *IEEE Trans. on Image Processing (T-IP)*, 30:1461-1475, 2021. (IF:11.041)
- **Gongyang Li**, Zhi Liu, Ran Shi, and Weijie Wei, “Constrained fixation point based segmentation via deep neural network,” *Neurocomputing*, 368:180-187, 2019. (IF:5.779)
- Ran Shi, **Gongyang Li**, Weijie Wei, and Zhi Liu, “Fixations based personal target objects segmentation,” *ACM Multimedia Asia*, 2020.
- Ran Shi, **Gongyang Li**, Weijie Wei, Xiaofei Zhou, and Zhi Liu, “Personalized image observation behavior learning in fixation based personalized salient object segmentation,” *Neurocomputing*, 445:255-266, 2021. (IF:5.779)

RGB-T/D Semantic Segmentation

- **Gongyang Li**, Yike Wang, Zhi Liu, Xinpeng Zhang, and Dan Zeng, “RGB-T semantic segmentation with location, activation, and sharpening,” *IEEE Trans. on Circuits and Systems for Video Technology (T-CSVT)*, 33(3):1223-1235, 2023. (IF: 5.859)
- Yike Wang[#], **Gongyang Li**[#], and Zhi Liu, “SGFNet: Semantic-guided fusion network for RGB-Thermal semantic segmentation,” *IEEE Trans. on Circuits and Systems for Video Technology (T-CSVT)*, 33(12):7737-7748, 2023. ([#] equal contribution) (IF: 5.859)
- Jianlin Chen, **Gongyang Li**[✉], Zhijiang Zhang[✉], and Dan Zeng, “EFDCNet: Encoding fusion and decoding correction network for RGB-D indoor semantic segmentation,” *Image and Vision Computing (IVC)*, 142:104892, 2024. (IF:3.860) ([✉] corresponding author)
- Ying Lv, Zhi Liu, and **Gongyang Li**, “Context-aware interaction network for RGB-T semantic segmentation,” *IEEE Trans. on Multimedia (T-MM)*, 26:6348-6360, 2024. (IF:8.182)

Segmentation-related

- **Gongyang Li**, Zhi Liu, and Xiaofei Zhou, “Effective online refinement for video object segmentation,” *Multimedia Tools and Applications*, 78(23):33617-33631, 2019. (IF:2.577)

- Zheng Hu, Zhi Liu, **Gongyang Li**, Linwei Ye, Lei Zhou, and Yang Wang, “Weakly supervised instance segmentation using multi-stage erasing refinement and saliency-guided proposals ordering,” *Journal of Visual Communication and Image Representation*, 73:102957, 2020. (IF:2.887)
- Yike Wang, Zhi Liu, **Gongyang Li**, Xiaofeng Lu, Xuefeng Liu, and Hongwei Zhang, “Hybrid modeling based semantic segmentation of forward-looking sonar images,” *IEEE Journal of Oceanic Engineering*, 50(1):380-393, 2025. (IF:3.8)
- Yuetong Li, Qing Zhang[✉], Yilin Zhao, Zeming Liu, and **Gongyang Li**[✉], “DGA-Net: Enhancing SAM with depth prompting and graph-anchor guidance for camouflaged object detection,” *IEEE Trans. on Circuits and Systems for Video Technology*, 2026.

Others

- Yong Wu[#], **Gongyang Li**[#], Zhi Liu, Mengke Huang, and Yang Wang, “Gaze estimation via modulation-based adaptive network with auxiliary self-learning,” *IEEE Trans. on Circuits and Systems for Video Technology (T-CSVT)*, 32(8):5510-5520, 2022. ([#] equal contribution) (IF: 5.859)
- Jing Lv, Zhi Liu[✉], and **Gongyang Li**[✉], “Few-shot fine-tuning with auxiliary tasks for video anomaly detection,” *Multimedia Systems (MS)*, 31:127, 2025. (IF:3.5) ([✉] corresponding author)
- Baojun Zhou, Xinpeng Huang, **Gongyang Li**, Chao Yang, Liquan Shen, and Ping An, “Dual-guided video frame interpolation with spatial-temporal global attention,” *IEEE Trans. on Multimedia (T-MM)*, 27:7783-7795, 2025.
- Shipei Wang, Ping An, Chao Yang, **Gongyang Li**, Xinpeng Huang, and Shiqi Wang, “Feature quality assessment: A database and a lightweight objective method,” *IEEE Trans. on Multimedia (T-MM)*, 27:7314-7325, 2025.
- Kaifan Zhao, Ran Ma, Min Su, Haozhe Yu, Ping An, and **Gongyang Li**, “MOFNet: Memory optimization fusion network for query-focused video summarization,” *IEEE Sensors Journal (SJ)*, 25(10):18334-18343, 2025.
- Ying Lv, Zhi Liu, **Gongyang Li**, and Xiaojun Chang, “Noise-aware intermediary fusion network for off-road freespace detection,” *IEEE Trans. on Intelligent Vehicles (T-IV)*, 2024.
- Qihan Jiao, Zhi Liu, **Gongyang Li**, Linwei Ye, and Yang Wang, “Fine-grained image classification with coarse and fine labels on one-shot learning,” *IEEE ICMEW*, 2020.
- Haocheng Gu, **Gongyang Li**, and Zhi Liu, “Masked feature regeneration based asymmetric student-teacher network for anomaly detection,” *Multimedia Tools and Applications*, 83:90573-90594, 2024. (IF:3)
- Yingjie Song, Zhi Liu, **Gongyang Li**, Jiawei Xie, Qiang Wu, Dan Zeng, Lihua Xu, Tianhong Zhang, and Jijun Wang, “EMS: A large-scale eye movement dataset, benchmark and new model for schizophrenia recognition,” *IEEE Trans. on Neural Networks and Learning Systems (T-NNLS)*, 36(5):9451-9462, 2025. (IF:10.2)
- Kaifan Zhao, Ran Ma, Haozhe Yu, Ping An, and **Gongyang Li**, “Query-focused video summarization driven by sentences,” *Displays*, 93:103398, 2026.
- Rui Gao, Meihong Zhang, **Gongyang Li**, Guanyi Li, Kai Zhao, Xianchao Zhang, and Dan Zeng, “MIST: A benchmark and baseline for multi-frame infrared small target detection in complex motion,” *IEEE Trans. on Image Processing (T-IP)*, 35:5196-5211, 2026.
- Haozhe Yu, Ran Ma, Shayan Yuan, Ping An, and **Gongyang Li**, “MAMF: Motion-aware multi-frame fusion framework for video stabilization,” *Displays*, 95:103623, 2026.
- Jing Lv, Zhi Liu, and **Gongyang Li**, “Video anomaly detection based on frequency-domain interaction and fusion,” *Displays*, 95:103628, 2026.

CONTRACTS AND GRANTS

- Gongyang Li. (12/2022) “Research on Visual Saliency Detection for Multi-source Images and Videos (2022253),” *Shanghai Super Postdoctoral Incentive Funding Program*, 300K RMB.

- Gongyang Li. (1/1/2023 - 6/30/2024) “Research on High-performance Saliency Detection for Complex Optical Remote Sensing Images (2022M722037),” *China Postdoctoral Science Foundation*, 80K RMB.
- Gongyang Li. (12/1/2024 - 11/30/2027) “Research on Weakly Supervised Saliency Detection for Multi-modal Data Based on Visual Segmentation Large Model (24YF2713000),” *Shanghai Rising Star Program (including Sailing Program)*, 200K RMB.
- Gongyang Li. (1/1/2025 - 12/31/2027) “Research on Saliency Detection for RGB-D Images Based on Depth Map Quality Perception (62401350),” *National Natural Science Foundation of China*, 300K RMB.
- Gongyang Li. (7/1/2024 - 6/30/2025) “Research on Multi-source Image Salient Object Detection Model based on Deep Learning (LERIS24-02),” *Opening Foundation of Quanzhou Vocational and Technical University in 2024*, 120K RMB.
- Gongyang Li. (9/1/2024 - 8/31/2026) “Research on Steel Surface Defect Detection Algorithm Based on Multimodal Data (KLMVI-2024-HIT-04),” *Open Research Fund of Anhui Province Key Laboratory of Machine Vision Detection and Perception*, 20K RMB.
- Gongyang Li. (1/1/2025 - 12/31/2025) “Research on Saliency Detection for Multi-modal Data Driven by Large Model (YNSC24109),” *Foundation of Yunnan Key Laboratory of Service Computing*, 10K RMB.
- Gongyang Li. (7/1/2025 - 12/31/2026) “Research on Multi-modal Image Salient Object Detection Based on Language and Visual Large Models (PRIU25-03),” *Foundation of Fujian Key Laboratory of Pattern Recognition and Image Understanding*, 50K RMB.

INVITED SPEECH

- 2025 9th International Conference on Advances in Image Processing, 11/7/2025 - 11/9/2025, Chengdu, *High-Performance and Lightweight Salient Object Detection Models in Optical Remote Sensing Images*.
- 2025 International Conference on Frontiers of Ocean Science and Technology, 11/21/2025 - 11/23/2025, Tianjin, *High-Performance and Salient Object Detection in Optical Remote Sensing Images*.

SERVICES

- Guest Editor for the Special Issue on *AI-Driven Image and Video Understanding* in *Journal of Imaging*.
- Guest Editor for the Special Issue on *Remote Sensing Data Dense Prediction and Feature Extraction* in *Remote Sensing*.
- Area Chair for IEEE MMSP2022.
- Journal Reviewers for IEEE T-PAMI, T-IP, T-VCG, T-NNLS, T-MM, T-CSVT, T-GRS, T-II, T-ITS, T-ETCI, T-BD, T-IM, T-CE, T-AI, IoT, JSTARS, Sensors Journal, SPL, GRSL, Sensors Letters, Journal of Photovoltaics; PR, Geo-spatial Information Science, Measurement, Neurocomputing, SPIC, JCST, SIVP, MTAP, JVC, Advanced Intelligent Systems, Applied Soft Computing Journal, IET Image Processing, European Journal of Remote Sensing, Journal of Intelligent & Fuzzy Systems, Electronic Research Archive, Journal of Real-Time Image Processing, Mathematical Biosciences and Engineering, and International Journal of Machine Learning and Cybernetics.
- Conference Reviewers for IEEE ICRA, IEEE ICASSP, IEEE MMSP, PRCV, IFTC.